# MQTT Broker

## Complete Guide

*localhost (Aedes) vs HiveMQ*

Project: IoT Project 2
Hardware: ESP8266 NodeMCU + Node-RED
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# 1. MQTT Kya Hai?

MQTT (Message Queuing Telemetry Transport) ek lightweight messaging protocol hai jo IoT devices ke liye design kiya gaya hai. Yeh Publish/Subscribe model pe kaam karta hai jisme devices data bhejte aur receive karte hain ek central 'Broker' ke through.

## 1.1 MQTT Architecture

| Component | Role | Hamare Project Mein |
|---|---|---|
| Publisher | Data bhejta hai | ESP8266 NodeMCU |
| Broker | Messages route karta hai | Aedes (local) / HiveMQ |
| Subscriber | Data receive karta hai | Node-RED Dashboard |
| Topic | Channel/address | tempNode, humiNode etc. |

## 1.2 Hamare Project ke MQTT Topics

| Topic Name | Direction | Data |
|---|---|---|
| tempNode | ESP → Node-RED | Temperature °C (e.g. 30.9) |
| humiNode | ESP → Node-RED | Humidity % (e.g. 26) |
| ldrNode | ESP → Node-RED | Light / Dark |
| mq135Node | ESP → Node-RED | Air Quality 0-1024 (ppm) |
| distanceNode | ESP → Node-RED | Distance in cm |
| led1StatusNode | ESP → Node-RED | ON / OFF |
| led2ControlNode | Node-RED → ESP | 1/ON or 0/OFF |

# 2. localhost — Aedes MQTT Broker

**Aedes Broker Kya Hai?**

Aedes ek Node-RED plugin hai jo directly Node-RED ke andar ek MQTT broker run karta hai. Iska matlab hai ki alag se koi software install karne ki zaroorat nahi — Node-RED khud hi broker ban jata hai!

## 2.1 Aedes Broker Kaise Kaam Karta Hai

Jab aap Node-RED mein Aedes broker node add karte hain, tab:

- Node-RED ek MQTT server start karta hai port 1883 pe
- Yeh server sirf local machine pe available hota hai
- ESP8266 PC ke hotspot IP se connect hota hai (e.g. 192.168.137.1)
- Node-RED MQTT nodes localhost se connect hote hain
- Dono same Aedes broker se baat karte hain

## 2.2 Network Diagram — Aedes (localhost)

```
        ESP8266 (192.168.137.109)
        |  WiFi Hotspot (192.168.137.1:1883)  |
          ↓                                  ↑
     PC — Aedes Broker (localhost:1883)
          ↓                                  ↑
   Node-RED MQTT Nodes (localhost:1883)
                         ↓
        Node-RED Dashboard (browser)
```

## 2.3 Aedes Setup — Step by Step

### Step 1: Plugin Install karo

Node-RED mein jao:

```
Menu (≡) → Manage Palette → Install Tab
Search: node-red-contrib-aedes
Install button click karo → Done
```

### Step 2: Aedes Broker Node Add karo

- Left panel mein 'aedes broker' dhundho
- Canvas pe drag karo
- Double click karo
- Name: localhost, MQTT Port: 1883
- Done → Deploy

### Step 3: MQTT IN Nodes Configure karo

Har MQTT IN node mein (Temperature, Humidity etc.):

- Server: 192.168.137.1 (PC Hotspot IP)
- Port: 1883
- Ya '+' button se naya broker add karo

### Step 4: ESP8266 Code

```
const char* mqtt_server = "192.168.137.1";  // PC Hotspot IP
const int mqtt_port = 1883;
```

## 2.4 Aedes ke Fayde aur Nuksan

| ☑ FAYDE (Advantages) | ✕ NUKSAN (Disadvantages) |
|---|---|
| Internet ki zaroorat nahi | Sirf local network mein kaam karta hai |
| Bahut fast — low latency | PC band ho to kaam nahi karta |
| Free — koi subscription nahi | Bahar se access nahi ho sakta |
| Alag software install nahi | Node-RED hamesha ON rakhna padta hai |
| Data private — bahar nahi jata | Multiple PC/location pe kaam nahi |
| Offline projects ke liye perfect | Production use ke liye suitable nahi |

## 2.5 Aedes Troubleshooting

| Problem | Solution |
|---|---|
| Nodes 'connecting' dikha rahe hain | MQTT node mein Server ko 192.168.137.1 karo (localhost nahi) |
| ESP connect nahi ho raha | Hotspot IP check karo — CMD mein 'ipconfig' chalao |
| Dashboard empty hai | Deploy dobara karo, topics match karo |
| Aedes node nahi dikh raha | node-red-contrib-aedes install karo Manage Palette se |

# 3. HiveMQ — Cloud MQTT Broker

**HiveMQ Kya Hai?**

HiveMQ ek public cloud MQTT broker hai jo internet pe available hai. Koi bhi free mein use kar sakta hai — koi login ki zaroorat nahi. Yeh globally accessible hai, matlab ESP8266 kisi bhi network se connect ho sakta hai.

## 3.1 HiveMQ Details

| Property | Value |
|---|---|
| Broker Address | broker.hivemq.com |
| MQTT Port | 1883 (non-secure) |
| MQTT Port (SSL) | 8883 (secure) |
| WebSocket Port | 8000 |
| Login Required | Nahi — completely anonymous |
| Cost | Free (public broker) |
| Message Limit | Unlimited (fair use) |
| Availability | 24/7 globally available |

## 3.2 Network Diagram — HiveMQ (Cloud)

```
ESP8266  ───── Internet ─────→ HiveMQ Cloud Broker
                               (broker.hivemq.com:1883)
                                                 |
Node-RED  ──── Internet ─────→ HiveMQ Cloud Broker
                               (broker.hivemq.com:1883)
                                                 |
          Dashboard (Browser — kahi se bhi!)
```

## 3.3 HiveMQ Setup — Step by Step

### ESP8266 Arduino Code

```
//const char* mqtt_server = "192.168.137.1";  // Local — comment out karo
const char* mqtt_server = "broker.hivemq.com";  // HiveMQ
const int mqtt_port = 1883;
```

### Node-RED MQTT Nodes Configure karo

- Har MQTT node pe double click karo
- Server dropdown mein 'HiveMQ' select karo

- Ya naya broker add karo: broker.hivemq.com, Port: 1883
- Deploy karo

## 3.4 HiveMQ ke Fayde aur Nuksan

| ☑ FAYDE (Advantages) | ✕ NUKSAN (Disadvantages) |
|---|---|
| Internet se kahi bhi access ho sakta | Internet connection must hai |
| Koi setup nahi — seedha use karo | Public broker — data secure nahi |
| Multiple devices worldwide connect ho sakte | Latency thodi zyada (internet delay) |
| PC hotspot ki zaroorat nahi | Koi bhi same topic subscribe kar sakta |
| Demo aur testing ke liye perfect | Server down ho to project band |
| Free mein cloud MQTT milta hai | Offline use nahi ho sakta |

# 4. Localhost (Aedes) vs HiveMQ — Full Comparison

| Feature | Aedes (localhost) | HiveMQ (Cloud) |
|---|---|---|
| Internet Required? | ❌ Nahi | ☑ Haan |
| Speed/Latency | 🚀 Bahut Fast (local) | ⚡ Medium (internet) |
| Setup Complexity | Medium (plugin install) | Easy (seedha use) |
| Remote Access | ❌ Sirf local | ☑ Worldwide |
| Data Privacy | ☑ 100% Private | ❌ Public broker |
| Reliability | PC pe depend karta hai | HiveMQ servers pe depend |
| Cost | Free | Free (public) |
| Multiple Networks | ❌ Ek network only | ☑ Any network |
| Best For | Home/Lab projects | Demo/Remote projects |
| Hamare Project Mein | ☑ Currently USE kar rahe | Pehle use kiya tha |

## 4.1 Kab Kya Use Karein?

**Aedes (localhost) use karo jab:**
• Ghar ya lab mein kaam kar rahe ho • Internet nahi ho ya slow ho • Data private rakhna ho • Fast response time chahiye • Offline demo deni ho

**HiveMQ use karo jab:**
• Kahin bhi remote access chahiye • Multiple locations pe devices hain • Quick demo ya testing karni ho • Internet hamesha available ho • Simple setup chahiye bina kisi plugin ke

# 5. Code Mein Switch Karna — Aedes ↔ HiveMQ

## 5.1 Arduino Code Switch

Sirf ek line change karni hai:

### HiveMQ ke liye:

```
const char* mqtt_server = "broker.hivemq.com";
// Koi WiFi hotspot zaroorat nahi
// ESP8266 kisi bhi WiFi se connect ho sakta
```

### Aedes (localhost) ke liye:

```
const char* mqtt_server = "192.168.137.1";  // PC Hotspot IP
// ESP8266 ko RSMaurya hotspot se connect karo
// PC pe Node-RED + Aedes chal raha hona chahiye
```

## 5.2 Node-RED Switch

Har MQTT node mein Server change karo:

| Setting | Aedes (local) | HiveMQ (cloud) |
|---|---|---|
| Server | 192.168.137.1 | broker.hivemq.com |
| Port | 1883 | 1883 |
| Username | (blank) | (blank) |
| Password | (blank) | (blank) |

# 6. Hamare Project ki Current Status

**Currently Chal Raha Hai:**
Aedes localhost broker chal raha hai aur ESP8266 se live data aa raha hai dashboard pe!

## 6.1 Working Configuration

| Parameter | Value |
|---|---|
| WiFi SSID | RSMaurya |
| ESP8266 IP | 192.168.137.109 |
| PC (Broker) IP | 192.168.137.1 |
| MQTT Broker | Aedes (Node-RED Plugin) |
| MQTT Port | 1883 |
| Dashboard URL | http://127.0.0.1:1880/ui |
| Node-RED URL | http://127.0.0.1:1880 |

## 6.2 Live Sensor Data (Tested Values)

| Sensor | Value | Unit | Status |
|---|---|---|---|
| Temperature | 31.1 | °C | ☑ Working |
| Humidity | 27 | % | ☑ Working |
| Air Quality | 80 | ppm | ☑ Working |
| Distance | 16.08 | cm | ☑ Working |
| LDR | Light | text | ☑ Working |
| LED2 Control | ON/OFF | switch | ☑ Working |

## 7. Author Information

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